

FishLog

Capstone Project — Design Document
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What is FishLog?

FishLog is a text-based C# console application for Ontario anglers. It lets the user log fishing trips and catches, and automatically checks whether each kept fish is legal under the 2026 Ontario Fishing Regulations before recording it. If a catch is illegal, the program explains why and records it as released instead.

What problem does it solve?

Ontario fishing regulations vary between zones and license types in ways that are easy to forget on the water. FishLog takes care of the rules so the angler doesn't have to. checking season dates, size limits, slot limits, and daily limits automatically based on the zone and license type the user sets up at the start of each trip.

Scope

The application covers two Fisheries Management Zones and six species:

- Zones: FMZ 10 and FMZ 11
- Species: Walleye, Northern Pike, Bass, Yellow Perch, Lake Trout, Muskellunge
- License types: Sport and Conservation
- Regulation data: 2026 Ontario Fishing Regulations Summary (effective January 1, 2026)

OOP Concepts Used

Concept	How it's used
Inheritance	License (abstract) → SportLicense, ConservationLicense. Species (abstract) → Walleye, NorthernPike, Bass, LakeTrout, Muskellunge, YellowPerch.
Polymorphism	ValidateCatch() is overridden by each species subclass with its own regulation logic. GetLimit() behaves differently per license type.
Encapsulation	All fields are private. Data is accessed through public methods only (e.g. GetLimit(), IsOpen(), PassesSizeCheck()).
Enums	LicenseType, FMZone, CatchStatus, and ValidationResult replace magic strings and numbers throughout the app.
Interfaces	ILoggable (GetSummary()) implemented by Angler, Trip, and Catch. IValidatable (Validate()) implemented by Trip and Catch.
Generics	TripLog<T> manages the angler's trip history. StatTracker<T> tracks catches and personal bests within a trip.
Error handling	All user input is wrapped in try/catch. Custom exceptions InvalidCatchException and InvalidInputException provide user-friendly messages.

Sample Interaction

```
Select zone: [1] FMZ 10  [2] FMZ 11 > 2
Species: [1] Walleye ... > 1
Length (cm): 52
Keep or release? [1] Keep  [2] Release > 1
```

```
ERROR: Walleye (52cm) falls within the protected slot (43-60cm) in FMZ 11.
Fish must be released.
```

UML Diagram

